

Vertex Separators and Graph Widths

Rafael Pérez

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Abstract

A classical result of Lipton and Tarjan states that n -vertex planar graphs can be separated into vertex sets of roughly equal size by removing $O(\sqrt{n})$ vertices. This result has been generalized in several directions. Moreover, it has been shown that separators that split a graph into roughly balanced vertex sets are closely tied to tree-width and path-width. In this paper we conduct a survey of some of these results, and extend the connection between separators and tree-width to directed graphs.

1 Introduction

All graphs in this paper are finite and simple, unless stated otherwise. Graphs in sections 2 and 3 are undirected, unless stated otherwise. Graphs in section 4 are directed, unless stated otherwise. For standard graph-theoretic definitions we refer the reader to [1]. For an undirected graph G , we say that $S \subseteq V(G)$ *separates* $A, B \subseteq V(G - S)$ if there is no path between a vertex in A and a vertex in B that does not go through a vertex in S . A *separation* of G is a pair (A, B) of vertex subsets of $V(G)$ such that $A \cap B$ separates $A \setminus B$ and $B \setminus A$, and $A \cup B = V(G)$. Its *order* is $|A \cap B|$.

A non-negative weight function $w : V(G) \rightarrow \mathbb{R}^+$ defined on the vertices is called a *weighting* of G . If G has a weighting w we say that it is *weighted*. If w assigns 1 to each vertex of G , we say that G is *unweighted*. For a vertex set $X \subseteq V(G)$ we let $w(X) := \sum_{v \in X} w(v)$.

Let G be a weighted graph with weighting w . For $\frac{1}{2} \leq \alpha < 1$ an α -*separator* of G of order k is a set $S \subseteq V(G)$ with $|S| = k$ such that there is a partition $A \cup B \cup S = V(G)$ with $w(A), w(B) \leq \alpha w(V(G))$ that separates A and B . A *balanced separator* of G is a set $S \subseteq V(G)$ such that each connected component $C \subseteq G - S$ satisfies $w(V(C)) \leq \frac{1}{2}w(V(G))$. We state the following connection between separators.

Lemma 1.1. *Let G be a weighted graph. If S is a balanced separator S of G then S is a $\frac{2}{3}$ -separator of G .*

Proof. Let G have weighting w . Let $C_1, \dots, C_n$ be the connected components of $G - S$. If there is some component C_i such that $w(V(C_i)) \geq \frac{1}{3}w(V(G))$ then let $A = V(C_i)$ and $B = V(C_1 \cup \dots \cup C_{i-1} \cup C_{i+1} \cup \dots \cup C_n)$. Otherwise, allocate connected components to the set A in order right until $w(A) \geq \frac{1}{3}w(V(G))$. Then, $w(A) < \frac{2}{3}w(V(G))$ since all components satisfy $w(V(C_i)) < \frac{1}{3}w(V(G))$. In both cases we clearly have $w(A), w(B) \leq \frac{2}{3}w(V(G))$. $\square$

Separators that split a graph into roughly balanced vertex sets are useful in many computational applications, especially if they can be found through an efficient algorithm and if their order is

relatively small. Namely, they find usefulness in many divide-and-conquer algorithms. Many of the results on separators are based on the topological structure of the graph. They are also closely linked to measures of width such as tree-width and path-width. We will also establish a link between separators with measures of width in directed graphs, which is an area that has not seen as much research as its undirected counterpart.

2 Separators According to Structure

In a directed tree T , we call the unique vertex that has all of its edges pointing away from it the *root* (if such a vertex exists). Suppose T has a root. For a vertex $t \in T$, we call the set of vertices with an edge pointing from v to them the *children* of t . Accordingly, t is the *parent* of these vertices. We call the vertices with a directed path from t to them the *descendants* of t (these include t itself). We let T_t be the subtree of T rooted at t (this is the tree that includes all the descendants of t).

Theorem 2.1. *Weighted trees have a balanced separator of order 1.*

Proof. Let T be a tree with n vertices and weights w . Define some vertex $r \in T$ as its root and direct the edges in T away from r . If every child c of r satisfies $w(V(T_c)) \leq \frac{1}{2}w(V(G))$ then we let $S = \{r\}$. Otherwise, there is some unique c such that $w(V(T_c)) > \frac{1}{2}w(V(G))$. We iterate this procedure on the children of c , redefining what c is, right until every child c' of c satisfies $w(V(T_{c'})) \leq \frac{1}{2}w(V(G))$. In this case we let $S = \{c\}$.

Observe that the component of $T - S$ which includes r has weight $< \frac{1}{2}w(V(G))$ since the subtree rooted at c has weight $> \frac{1}{2}w(V(G))$. By construction, the other components of $T - S$ have weight $\leq \frac{1}{2}w(V(G))$, so we are done. $\square$

The following then follows directly from lemma 1.1.

Theorem 2.2. *Weighted trees have a $\frac{2}{3}$ -separator of order 1.*

Having found $\frac{2}{3}$ -separators for trees of constant order, it stands to reason that $\frac{1}{2}$ -separators for trees should also be fairly small. We will show that trees have $\frac{1}{2}$ -separators of order $\Theta(\log n)$, where n denotes the number of vertices. To show that they have $\frac{1}{2}$ -separators of order $O(\log n)$ it is convenient to prove the following, more general lemma which was first shown (albeit less generally) by Lipton and Tarjan [2].

Lemma 2.3. *Let G be a graph with n vertices such that all unweighted subgraphs have an α -separator of order $\leq \beta f(n)$, where $f : \mathbb{N} \rightarrow \mathbb{R}^+$. Then for every weighting of G there exists $\frac{1}{2}$ -separator of G order $\leq (\lceil \log_{1/\alpha}(n) \rceil + 1)f(n)$. Furthermore, if $f(n) = \beta n^c$ for some $\beta > 0$ and $0 < c < 1$ then the order is $\leq f(n)/(1 - \alpha^c)$.*

Proof. Fix a weighting w of G . Define the sequences of vertex sets $\{A_i\}, \{B_i\}, \{C_i\}, \{D_i\}$ such that:

1. A_i, B_i, C_i, D_i partition $V(G)$.
2. No edge joins A_i with B_i , A_i with D_i , or B_i with D_i .
3. $w(A_i) \leq w(B_i)$ and $w(B_i) \leq w(A_i \cup C_i \cup D_i)$.
4. $|D_i| \leq \alpha |D_{i-1}|$

Let $A_0 = B_0 = C_0 = \emptyset, D_0 = V$. Then 1 – 4 hold. Now, if $A_{i-1}, B_{i-1}, C_{i-1}, D_{i-1}$ are defined and $D_{i-1} \neq \emptyset$ let G' be the subgraph of G induced by D_{i-1} . Let C' be an α -separation of the unweighted G' into the partition A', B', C' . This means that $|A'|, |B'| \leq \alpha |D_{i-1}|$

Without loss of generality, suppose $w(A') \leq w(B')$. Let A_i be the set among $A_{i-1} \cup A', B_{i-1}$ with lower weight and let B_i be the other set. Let $C_i = C_{i-1} \cup C'$ and let $D_i = B'$. Then 1 – 4 hold for A_i, B_i, C_i, D_i .

Let k be the largest index for which A_k, B_k, C_k, D_k are defined. Then $D_k = \emptyset$. Let $A = A_k, B = B_k, C = C_k$. By 1 and 2, C separates A, B . By 3, $w(A), w(B) \leq \frac{1}{2}w(V(G))$. By 4, the total number of vertices in C is bounded by

$$\sum_{i=0}^{\lfloor \log_{1/\alpha}(n) \rfloor + 1} f(\alpha^i n) \leq (\lfloor \log_{1/\alpha}(n) \rfloor + 1)f(n)$$

If $f(n) = \beta n^c$ then we can get the asymptotically better bound

$$\sum_{i=0}^{\lfloor \log_{1/\alpha}(n) \rfloor + 1} f(\alpha^i n) = \beta n^c \sum_{i=0}^{\lfloor \log_{1/\alpha}(n) \rfloor + 1} \alpha^{ic} \leq \beta n^c \frac{1}{1 - \alpha^c}$$

□

We say that a tree T is an m -ary tree if we can direct the edges such that one vertex is the root, and each vertex has at most m children. We say that the *depth* of a vertex is the length of the path from the root. We say that the *height* of such a tree is the length of the longest path from the root to a leaf. We say that an m -ary tree is *complete* if all its vertices have 0 or m children and all leafs have the same height. A *rooted binary tree* is a 2-ary tree and a *rooted ternary tree* is a 3-ary tree.

It turns out that the result of lemma 2.3 is tight. The following was shown by Bollobás, Chung and Graham [3].

Theorem 2.4. *Let T be an unweighted rooted ternary tree with height $2h$ and $n = (3^{2h} - 1)/2$ vertices. Then T has no $\frac{1}{2}$ -separator of order less than $\frac{\log n}{2 \log 3} - \frac{\log n}{\log \log n}$.*

The proof of this statement is fairly complicated. We give a weaker result that has the advantage of being shorter and easier to follow. The proof of the statement also relies on a different strategy.

Theorem 2.5. *There exist arbitrarily large weighted rooted binary trees with n vertices such that there is no $\frac{1}{2}$ -separator of order less than $\Omega(\log_3 \log_2 n)$.*

Proof. Let T be a complete rooted binary tree with $n = 2^{3^k+1} - 1$ vertices for $k \in \mathbb{N}$ and a weight function w that is 0 on all non-leaf vertices. On a leaf vertex v , we let $w(v) = (\frac{2}{3})^r (\frac{1}{3})^\ell$ where starting from the root, r is the number of times one has to choose the right path and ℓ is the number of times one has to choose the left path in order to get to v . Let $h = 3^k$ be the height of T . Note that for leaf vertices we have $r + \ell = h$.

Observe that all $\frac{1}{2}$ -separators of T must include a vertex. This follows from the simple observation that there is no way to sum a finite number of powers of $\frac{1}{3}$ and $\frac{2}{3}$ to get exactly $\frac{1}{2}$.

Let $T^0 = T$. For $i \geq 1$, we define a sequence T^i of complete rooted binary trees sequentially such that each complete rooted binary tree has one third the height as the previous one. The weights of

the leaves of T^i are the same as the weights that the descendants of the corresponding vertices at T^{i-1} had. However, at each tree T^i we remove one leaf vertex.

Observe that for each i , if we determine that we need to remove a leaf vertex from T^i in order to obtain a $\frac{1}{2}$ -separator, then we would have also needed to remove at least one vertex from T from vertices at height 3^i or lower in order to obtain a $\frac{1}{2}$ -separator. This is because when we remove a leaf vertex from T^i , we are removing all the weight contributions of all of the descendants of the corresponding vertex in T . Indeed, if a $\frac{1}{2}$ -separator S of T includes a vertex t , then $S \cup T_t$ is also a $\frac{1}{2}$ -separator of T .

Now, we determine up until which i we are forced to remove a leaf vertex from T^i . Indeed, suppose we have removed a leaf vertex at T^{i-1} and consider the resulting tree T^i . Then the leaf vertices have weights according to $(1/3)^\ell (2/3)^r$ for some $\ell + r = \frac{h}{3^i}$, except that, collectively, they have had weight subtracted of at most

$$\sum_{j=0}^{i-1} \left(\frac{2}{3}\right)^{\frac{h}{3^j}} \underbrace{\leq}_{\text{if } h \geq 3^{i-1}} 2 \cdot \left(\frac{2}{3}\right)^{\frac{h}{3^{i-1}}}$$

(this sum represents the removal of the largest-weight leaf vertex in all trees in the list up to T^i). Now, we consider whether we can get a $\frac{1}{2}$ -separator by just removing non-leaf vertices on T^i . Suppose we were to partition components of T^i freely to vertex sets A and B without removing a single leaf vertex. Suppose without loss of generality that $w(A) \geq w(B)$, so it suffices to prove we can achieve $w(A) \leq \frac{1}{2}$. Since the denominator of the weight of the leaf vertices is $3^{\frac{h}{3^i}}$ then the closest we can get to this is

$$w(A) - \frac{1}{2} \geq \frac{1}{2 \cdot 3^{\frac{h}{3^i}}} - 2 \cdot \left(\frac{2}{3}\right)^{\frac{h}{3^{i-1}}} = \frac{3^{\frac{2h}{3^i}} - 2^{\frac{h}{3^{i-1}}+2}}{2 \cdot 3^{\frac{h}{3^{i-1}}}}$$

(the first term follows from the denominator of $3^{\frac{h}{3^i}}$ and the second follows from the weight that was subtracted).

$$\geq \frac{1}{2 \cdot 3^{\frac{h}{3^i}}} - 2 \cdot \left(\frac{2}{3}\right)^{\frac{h}{3^{i-1}}} = \frac{3^{\frac{2h}{3^i}} - 2^{\frac{h}{3^{i-1}}+2}}{2 \cdot 3^{\frac{h}{3^{i-1}}}}$$

It suffices to determine when $3^{\frac{2h}{3^i}} - 2^{\frac{h}{3^{i-1}}+2} \leq 0$ to get a lower bound on i . We rearrange and take logs to get

$$\begin{aligned} \frac{2h}{3^i} \ln(3) &\leq \left(\frac{h}{3^{i-1}} + 2\right) \ln(2) \\ \iff \frac{2h \ln(3) - 3h \ln(2)}{2 \ln(2)} &\leq 3^i \\ \iff \log_3(h) + \log_3\left(\frac{2 \ln(3) - 3 \ln(2)}{2 \ln(2)}\right) &\leq i \end{aligned}$$

we express this in terms of n by substituting $h = \log_2(n+1) - 1$ so

$$\iff \log_3(\log_2(n+1) - 1) + \underbrace{\log_3\left(\frac{2 \ln(3) - 3 \ln(2)}{2 \ln(2)}\right)}_{>0} \leq i$$

Hence, $i = \Omega(\log_3 \log_2(n))$. □

It is not difficult to get similar results as in theorems 2.1 and 2.2 for outerplanar graphs. Let S be a surface. A *line* in S is a subset homeomorphic to $[0, 1]$. A *drawing* on a surface S is a graph G such that $V(G) \subseteq S$ and each $e = uv \in E(G)$ is a line with ends $u \neq v$ in $V(G)$. A graph G is *planar* if it can be *drawn* on the plane such that no two edges intersect, except possibly at the ends, and a vertex and edge intersect if and only if they are incident. The vertices and edges of a planar graph split the plane into different regions called *faces*. The *outer face* is the only unbounded region, not enclosed by vertices or edges. A graph G is *outerplanar* if it can be drawn on the plane such that all vertices are incident on the outer face.

Theorem 2.6. *Weighted outerplanar graphs have a $\frac{2}{3}$ -separator of order 2.*

Proof. Let G be an outerplanar graph with weighting w . We may assume $|V(G)| > 3$ since otherwise the theorem follows trivially. If some vertex $v \in G$ satisfies $w(v) \geq \frac{1}{3}w(V(G))$ then $S = \{v\}$ is a $\frac{2}{3}$ -separator, so the theorem follows. So we may assume $w(v) < \frac{1}{3}w(V(G))$ for all vertices v .

Let G' be a maximal outerplanar graph of G (i.e. add as many edges as possible between vertices in G such that the graph is still simple and outerplanar). Observe that G is 2-connected and all faces, except the outer one, are triangles. Clearly, a $\frac{2}{3}$ -separator of G' is also a $\frac{2}{3}$ -separator of G . Draw G' in the plane. We consider the following procedure.

Pick an arbitrary facial cycle composed of vertices u, v, w such that they appear in this order as you go in a clockwise direction along the outer face of G' . Without loss of generality, let u, v be the vertices that are not next to each other if we go along the outer face of G' . It is easy to see that the removal of u, v separates G' into exactly two components C_1, C_2 . If both the components obtained have weight $\leq \frac{2}{3}w(V(G))$ then $S = \{u, v\}$ is our desired separator.

Otherwise, suppose without loss of generality that $w(C_1) > \frac{2}{3}w(V(G))$. Consider the triangle in G' adjacent to the edge uv . Then u, v, z are the vertices of this triangle for some vertex $z \neq w$. If removing u, z results in a graph that is still connected, and removing v, z also results in a graph that is still connected, then uz and vz are both incident on the outer face. Hence, $C_1 = \{z\}$ and so $w(z) > \frac{2}{3}w(V(G))$, which contradicts our assumption that all vertices have weight at most $\frac{1}{3}w(V(G))$.

So, and without loss of generality, suppose removing v, z separates G' into components C'_1, C'_2 . Observe that the component, say C'_2 , containing u satisfies $C'_2 = C_2 \cup u$ and hence

$$w(C'_2) = w(C_2) + w(u) < \frac{1}{3}w(V(G)) + \frac{1}{3}w(V(G)) = \frac{2}{3}w(V(G))$$

If $w(C'_1) \leq \frac{2}{3}w(V(G))$ then letting $S = \{v, z\}$ concludes the proof. Otherwise, $w(C'_1) > \frac{2}{3}w(V(G))$ and we continue this procedure until both components have weight at most $\frac{2}{3}w(V(G))$.

Observe that this procedure terminates since $|C'_1| < |C_1|$. Indeed, at each step, the component with weight $> \frac{2}{3}w(V(G))$ loses a vertex. $\square$

We remark that, unlike trees, it is not possible to obtain a balanced separator of order 2 in general. However, a balanced separator of order 3 is possible. To see this, we define the *dual* G^* of a planar graph G , which is obtained by letting the vertices be the faces of G , which have an edge between them if the corresponding faces share an edge. If we let G' be the maximal outerplanar graph of G and $(G')^*$ be its dual then it is not hard to see that $(G')^*$ is a tree. So we can apply theorem 2.1 to get the result.

Theorem 2.7. *Weighted outerplanar graphs have a balanced separator of order 3.*

The following well-known theorem was proven by Lipton and Tarjan [2] and is one of the earliest known vertex separator theorems.

Theorem 2.8 (Planar Separator Theorem). *Let G be a weighted planar graph with n vertices. Then there exists a $\frac{2}{3}$ -separator of G of order $\leq 2\sqrt{2}\sqrt{n}$.*

The authors also give an algorithm that can find such separator in $O(n)$ time. The asymptotic tightness of theorem 2.8 can be easily verified by considering a *square grid graph*, which consists of k rows and k columns of vertices arranged straight lines with edges connecting neighboring vertices to the north, south, east and west of each. We will first need the following lemma, shown by Lipton and Tarjan [2].

Lemma 2.9. *Let G be the $k \times k$ square grid graph and let $A \subseteq V(G)$ satisfy $\alpha n \leq |A| \leq n/2$ for $n = k^2$ and $0 < \alpha < \frac{1}{2}$. Then the number of vertices in $G - A$ adjacent to some vertex in A is at least $k \cdot \min\{1/2, \alpha\}$.*

Theorem 2.10. *Let G be the unweighted $k \times k$ square grid, where $k \geq 6$. Then G has no $\frac{2}{3}$ -separator of order less than $k/2$.*

Proof. Let S be a $\frac{2}{3}$ -separator of G , partitioning $V(G)$ into A, B, S . Without loss of generality suppose $|A| \leq n/2$. Observe that we require $n/3 \leq |A \cup S|$. Suppose for contradiction that $|S| < k/2$. Then $k^2/3 - k/2 \leq |A|$, and since $k \geq 6$ then we get

$$k^2/3 - k/2 = \frac{k^2(k - \frac{3}{2})}{3} \geq \frac{k(3k/4)}{3} = \frac{n}{4}$$

so $n/4 \leq |A|$ and by lemma 2.9 we get that the number of vertices in $G - A$ adjacent to some vertex in A is at least $k/2$. But this contradicts the fact that $|S| < k/2$ since this vertex set of this size would not be able to separate A and B . Hence, $|S| \geq k/2$. $\square$

We remark that if we let $n = k^2$, then the best bound for this result approaches $\sqrt{2n/3}$ as n approaches infinity. The bound in theorem 2.10 was chosen for convenience.

The best known lower bound on the constant for $\frac{2}{3}$ -separators is $\sqrt{4\pi\sqrt{3}}/3$ and was obtained by Djidjev [4]. The best known constant for unweighted $\frac{2}{3}$ -separators is 1.97 while the best for $\frac{2}{3}$ -separators with weights is 2. Both of these were proven by Djidjev and Venkatesan [5].

It turns out that we can generalize the planar separator theorem in multiple ways. A graph has *genus g* if it can be drawn on a surface of genus g such that no two edges intersect, except possibly at their ends, and a vertex and edge intersect if and only if they are incident. Gilbert, Hutchinson and Tarjan [6] showed the following.

Theorem 2.11. *Let G be a weighted graph of genus $g > 0$ with n vertices. Then there exists a $\frac{2}{3}$ -separator of G of order $\leq 6\sqrt{gn} + 2\sqrt{2n} + 1$.*

They also provided an $O(n + g)$ algorithm to find said separator, given an embedding of G on a surface of genus g . However, finding an embedding of G on a surface of genus g is NP-hard when g is part of the input, which limits its applicability.

Alon, Seymour and Thomas [7] proved the following result based on a much more general condition. We call K_h the *complete* graph on h vertices, where each pair of vertices has an edge between them. It is possible to show that if a graph G has no K_h minor then its genus is $\leq O(h^2)$.

Theorem 2.12. *Let G be a weighted graph without K_h as a minor, with n vertices. Then there exists a $\frac{2}{3}$ -separator of G of order $\leq O(h^{3/2}\sqrt{n})$.*

They also provided an algorithm with running time $O(\sqrt{h}\sqrt{nm})$ where $m = n + |E(G)|$ to find said separator. The authors also conjectured that a $\frac{2}{3}$ -separator of order $\leq O(h\sqrt{n})$ should be possible, which would generalize theorem 2.11. This was proven true by Kawarabayashi and Reed [8].

Theorem 2.13. *Let G be a weighted graph without K_h as a minor, with n vertices. Then there exists a $\frac{2}{3}$ -separator of G of order $\leq O(h\sqrt{n})$.*

Recently, Bonnet, et al. [9] provided an algorithm with running time $O(\text{poly}(h)n)$ for a $\frac{2}{3}$ -separator of G of order $\leq O(\text{poly}(h)\sqrt{n})$, giving the first linear-time algorithm for a separator of square root order (when taking only n as input).

An *intersection graph* is a graph formed from sets S_i , $0 \leq i \leq n$ by creating one vertex v_i for each S_i and connecting v_i and v_j by an edge if $S_i \cap S_j \neq \emptyset$. A *string graph* is the intersection graph of curves in the plane. Fox and Pach [10] were the first to prove a separator theorem for string graphs.

Theorem 2.14. *Let G be a weighted string graph with m edges. Then there exists a $\frac{2}{3}$ -separator of G of order $\leq O(m^{3/4}\sqrt{\log m})$.*

In their paper, they conjectured the existence of a $\frac{2}{3}$ -separator of order $\leq O(\sqrt{m})$. Matoušek [11] improved the order to $\leq O(\sqrt{m} \log m)$. Lee [12] proved the following stronger result, which also implies the conjecture of Fox and Pach.

A graph G is a *region intersection graph* over G_0 if G is an intersection graph of a set of connected subsets of G_0 .

Theorem 2.15. *Let G be a weighted rig over G_0 where G_0 excludes K_h as a minor and G has m edges. Then there exists a $\frac{2}{3}$ -separator of G of order $\leq O(h^3\sqrt{\log h}\sqrt{m})$.*

It can be shown that string graphs are precisely the region intersection graphs over planar graphs, so this theorem implies the conjecture (since if a graph excludes K_6 then it is a planar graph).

3 Tree-width and Path-width

A *tree-decomposition* $\mathcal{T} = (T, X_t)$ of a graph G is a tree T and family X_t of subsets of $V(G)$ where $t \in T$ such that

(T1) $V(G) = \bigcup_{t \in T} X_t$.

(T2) Every edge of G has both ends in some X_t for $t \in T$.

(T3) For $t_1, t_2, t_3 \in T$, if t_2 lies in the path in T from t_1 to t_3 then $X_{t_1} \cap X_{t_3} \subseteq X_{t_2}$.

Similarly, a *path-decomposition* of a graph G is a sequence $X_1, \dots, X_r$ of subsets of $V(G)$ such that

(P1) $V(G) = \bigcup_{1 \leq i \leq r} X_i$.

(P2) Every edge of G has both ends in some X_i for $1 \leq i \leq r$.

(P3) For $1 \leq i_1 \leq i_2 \leq i_3 \leq r$, $X_{i_1} \cap X_{i_3} \subseteq X_{i_2}$

In the notation of both tree-decompositions and path-decompositions, we call each X_i a *bag*. Observe that a path-decomposition is a type of tree-decomposition since every path-decomposition is a tree-decomposition where T is a path.

The *tree-width* of G is the minimum over all tree-decompositions $\mathcal{T} = (T, X_t)$ of $\max_{t \in T} |X_t| - 1$. We denote this value by $\text{tw}(G)$. Similarly, the *path-width* of G is the minimum over all path-decompositions $(X_1, \dots, X_r)$ of $\max_{1 \leq i \leq r} |X_i| - 1$. We denote this value by $\text{pw}(G)$.

It is not hard to see that the tree-width of a tree is 1, and the tree-width of an outerplanar graph is 2. Similarly, it is not hard to see that the path-width of paths is 1 and the path-width of cycles is 2. By the observation made previously, it is easy to see that $\text{tw}(G) \leq \text{pw}(G)$. Finally, it is well-known that tree-width and path-width are closed under the minor relation.

Theorem 3.1. *Let G be a graph and H be a minor of G . Then $\text{tw}(H) \leq \text{tw}(G)$ and $\text{pw}(H) \leq \text{pw}(G)$.*

3.1 Separators and Tree-width

Suppose we have a tree-decomposition $\mathcal{T} = (T, X_t)$ of G . For each vertex $t \in T$ we call the connected components of $T - t$ the *branches* of T at t . We will need the following easy-to-prove lemmas [13].

Lemma 3.2. *For $t \in T$ and $v \in V(G)$ either $v \in X_t$ or there is a branch of T at t that contains all $t' \in T$ such that $v \in X_{t'}$.*

If $v \notin X_t$ then let $B_t(v)$ denote the branch described in lemma 3.1. This branch is unique and well-defined by the lemma.

Lemma 3.3. *If $v, v' \notin X_t$ are not separated in G by X_t then $B_t(v) = B_t(v')$.*

We have the following theorems that establish a linear dependence between tree-width and size of a balanced separator. The first one was first proven (albeit with less generality) by Seymour and Robertson [13].

Theorem 3.4. *Let G be a weighted graph. If $\text{tw}(G) < k$ then there exists a balanced separator of G of order $\leq k$.*

Proof. Let G have weighting w . Let $\mathcal{T} = (T, X_t)$ be a tree-decomposition of G with $|X_t| \leq k$ for all $t \in T$. Define some vertex $r \in T$ as its root and direct the edges in T away from r . We abuse notation and for each $t \in T$, let $X_{T_t} := \bigcup_{t' \in T_t} X_{t'}$. Observe that if p is the parent of t then T_t is a branch of T at p . We establish the following algorithm.

If r has the property that for each of its children c we have

$$w(X_{T_c}) \leq \frac{1}{2}w(V(G))$$

then we can let $S := X_r$ be the balanced separator. Indeed, we clearly have $|S| \leq k$. Moreover, for each component $C \subseteq G - S$, all vertices $v \in C$ are part of the same branch $B_r(v)$ by lemma 3.3. Since all branches of T at r are equal to T_c for some c then $w(V(C)) \leq \frac{1}{2}w(V(G))$.

Otherwise, there is some child c of r such that

$$w(X_{T_c}) > \frac{1}{2}w(V(G))$$

Continue this algorithm on the children of c , redefining what c is in each iteration, right until every child c' of c satisfies

$$w(X_{T_{c'}}) \leq \frac{1}{2}w(V(G))$$

which must eventually occur since eventually we will reach a c that is a leaf vertex (in which case the above is vacuously true since a leaf has no children).

In this case we can let $S := X_c$ be the balanced separator. Indeed, we clearly have $|S| \leq k$. Let B be the branch of T at c that contains the root r . It suffices to check that

$$w(X_B \setminus X_c) \leq \frac{1}{2}w(V(G))$$

Observe that by our choice of when the algorithm stops then we have $w(X_{T_c}) > \frac{1}{2}w(V(G))$. Hence, $w(V(G - X_{T_c})) < \frac{1}{2}w(V(G))$. Moreover, by the contrapositive of lemma 3.3, X_c separates $B \setminus X_c$ and $X_{T_c} \setminus X_c$. So we have

$$V(G - X_{T_c}) \supseteq B \setminus X_{T_c} = B \setminus X_c$$

Thus, $w(X_B \setminus X_c) < \frac{1}{2}w(V(G))$.

Finally, for each component $C \subseteq G - S$, all vertices $v \in C$ are part of the same branch $B_c(v)$ by lemma 3.3. Since all branches of T at c are either equal to B or to $T_{c'}$ for some c' then $w(V(C)) \leq \frac{1}{2}w(V(G))$. $\square$

The following theorem is well-known and commonly attributed to Robertson and Seymour [13], although we have been unable to find a direct proof by them. The following proof is partly due to Houdrouge, Miraftab and Morin [14]. We say that G is a graph with 0-1 *weighting* if its associated weight function is of the form $w : V(G) \rightarrow \{0, 1\}$.

Theorem 3.5. *Let G be a graph such that for every 0-1 weighting of G there exists a $\frac{2}{3}$ -separator of order $\leq k$. Then $\text{tw}(G) < 4k$.*

Proof. We define the following recursive algorithm. It takes as input a graph G , a separation (X, Y) with $W := X \cap Y$ where $|W| \leq 3k$, and a tree-decomposition $\mathcal{T} = (T, X_t)$ of $G[Y]$ of width at most $4k$ in which some bag X_t contains all the vertices in W .

If $|X| \leq 3k$ then we can create a leaf t' in T adjacent to t with $X_{t'} := X$ and we are done. So we may assume $|X| \geq 3k + 1$. Since $|W| < 3k + 1$ then we can take vertices from $X \setminus Y$ and add them to W to create a new vertex set W^* , where $|W^*| = 3k + 1$.

To extend $\mathcal{T}$, we let G have weighting w which sets the vertices of W^* to 1 and all other vertices in G to 0. Let (A', B') be a separation of G such that $S' := A' \cap B'$ is a $\frac{2}{3}$ -separator of G of order $\leq k$. Let $(A, B) := (A' \cap X, B' \cap X)$ and let $G[X]$ have weighting w (where w is restricted to the vertices in X). Then (A, B) is a separation of $G[X]$ such that $S := A \cap B$ is a $\frac{2}{3}$ -separator of $G[X]$ of order $\leq k$. Create a leaf t' in T adjacent to t with $X_{t'} := W \cup S$. Hence, $|X_{t'}| \leq 4k$ so the width of our tree-decomposition is still less than $4k$.

Let $X_A := A, Y_A := Y \cup S$ and $G_A := G[X_A \cup Y_A]$. What we get is a tree-decomposition of $G[Y_A]$ and a separation (X_A, Y_A) of G_A , with $W_A = X_A \cap Y_A$. Observe that $|W_A| \leq |W \setminus B| + |S| \leq |W^* \setminus B| + |S| \leq \frac{2}{3}|W^*| + k < 3k + 1$. Moreover, a bag $X_{t'}$ contains $W_A \subseteq W \cup S$. So we can call the algorithm on $G_A, (X_A, Y_A), \mathcal{T}$. This way, the algorithm extends $\mathcal{T}$ to G_A .

Now, we let $X_B := B, Y_B := Y \cup A$ and $G_B := G[X_A \cup Y_A] = G$. What we get is a tree-decomposition of $G[Y_B]$ and a partition (X_B, Y_B) of G_B , with $W_B = X_B \cap Y_B$. Observe that

$|W_B| \leq |W \setminus A| + |S| \leq |W^* \setminus A| + |S| \leq \frac{2}{3}|W^*| + k < 3k + 1$. Moreover, a bag $X_{t'}$ contains $W_B \subseteq W \cup S$. So we can call the algorithm on $G_B, (X_B, Y_B), \mathcal{T}$. This way, the algorithm extends $\mathcal{T}$ to $G_B = G$.

To initialize this algorithm, we let $X = V(G)$ and Y be some set of $3k$ vertices, and let $\mathcal{T} = (T, X_t)$ be a tree with one vertex t and bag $X_t = Y$ (if $|V(G)| < 3k$ then we get our desired tree-decomposition trivially). Observe that this algorithm terminates. This is because at each step it either doesn't recurse or we have $|W^*| = 3k + 1$ and $|S| \leq k$, meaning that $A, B \subsetneq X$ (indeed if, say, $A = X$ then $w(A \setminus B) = w(X) - w(S) \geq 2k + 1 > \frac{2}{3}(3k + 1) = \frac{2}{3}w(X)$ which contradicts the fact that S is a $\frac{2}{3}$ -separator of $G[X]$). This means that each time we call the algorithm recursively, X gets smaller, and so we will eventually get $|X| \leq 3k$ which is the stopping condition of the recursion. $\square$

We remark that this result may be generalized to α -separators for every 0-1 weighting by maintaining a separation of order $\leq \frac{1}{1-\alpha}k$ to obtain tree-width less than $\frac{2-\alpha}{1-\alpha}k$. So if $\alpha = \frac{1}{2}$ then $\text{tw}(G) < 3k$.

Dvořák and Norin [15] proved a substantial strengthening of theorem 3.5, although the proof is much more complicated.

Theorem 3.6. *If every unweighted subgraph of G has a $\frac{2}{3}$ -separator of order $\leq k$ then $\text{tw}(G) \leq 15k$.*

3.2 Separators and Path-width

As with tree-width, it is possible to bound the order of α -separators using path-width. However, we obtain much weaker results in the converse.

Theorem 3.7. *Let G be a weighted graph. If $\text{pw}(G) < k$ then there exists a $\frac{1}{2}$ -separator of G of order $\leq k$.*

Proof. Let G have weighting w . Let X_i for $1 \leq i \leq r$ be a path-decomposition of G as given by lemma. If $r = 1$ then we can let $S = X_1$ and we are done. So we may assume $r \geq 2$. Let $1 \leq k < r$ be the largest integer such that

$$w\left(\bigcup_{i=1}^k X_i\right) \leq \frac{1}{2}w(V(G))$$

If no such k exists then $w(X_1) > \frac{1}{2}w(V(G))$ so we can let $S = X_1$ and we are done. Observe that by the maximality of k ,

$$w\left(\bigcup_{i=1}^{k+1} X_i\right) > \frac{1}{2}w(V(G))$$

and hence,

$$w\left(\bigcup_{i=k+2}^r X_i \setminus X_{k+1}\right) < \frac{1}{2}w(V(G))$$

(by (P3), $\bigcup_{i=1}^{k+1} X_i$ and $\bigcup_{i=k+2}^r X_i \setminus X_{k+1}$ are disjoint). So if we let

$$S = X_{k+1}, \quad A = \bigcup_{i=1}^k X_i \setminus S, \quad B = \bigcup_{i=k+2}^r X_i \setminus S$$

then clearly, A, B, S is a partition of $V(G)$ where $w(A), w(B) \leq \frac{1}{2}w(V(G))$ and $|S| \leq k$. Moreover, S separates A and B by lemma 3.3 (since a path-decomposition is a tree-decomposition over a path P on vertices $\{1, \dots, r\}$, and $\{1, \dots, k\}$ and $\{k+2, \dots, r\}$ are the branches of P at $k+1$). $\square$

Korach and Solel [16] showed that if G has n vertices then $\text{pw}(G) = O(\text{tw}(G) \log n)$ by showing that forests have path-width of $O(\log n)$. The following result is asymptotically equivalent, but proven directly using α -separators.

Theorem 3.8. *Let G be a graph with n vertices such that every unweighted subgraph of G has an α -separator of order at most $f(n)$. Then $\text{pw}(G) \leq (\lfloor \log_{1/\alpha}(n) \rfloor + 1)f(n)$. Furthermore, if $f(n) = \beta n^c$ for some $\beta > 0$ and $0 < c < 1$ then $\text{pw}(G) \leq f(n)/(1 - \alpha^c)$.*

Proof. We consider the following recursive algorithm. The superscript denotes the depth of the recursion.

Partition G into A_1^1, B_1^1, S_1^1 , where $|A_1^1|, |B_1^1| \leq \alpha n$ and $|S_1^1| \leq f(n)$. Now, let $(X_1, \dots, X_r)$ and $(Y_1, \dots, Y_q)$ be path-decompositions of $G[A_1^1]$ and $G[B_1^1]$ respectively, each of which have path-width $\leq \alpha n - 1$ (indeed, we can let each path-decomposition be one bag containing A_1^1 and B_1^1). Then $(X_1 \cup S_1^1, \dots, X_r \cup S_1^1, Y_1 \cup S_1^1, \dots, Y_q \cup S_1^1)$ is a path-decomposition of G with path-width $\leq \alpha n + f(n) - 1$.

We now partition $G[A_1^1]$ into A_1^2, B_1^2, S_1^2 where $|A_1^2|, |B_1^2| \leq \alpha^2 n$ and $|S_1^2| \leq f(\alpha n)$. We let $(X_1, \dots, X_r)$ and $(Y_1, \dots, Y_q)$ be path-decompositions of $G[A_1^2]$ and $G[B_1^2]$ respectively, each of which have path-width $\leq \alpha^2 n - 1$. Then $(X_1 \cup S_1^2, \dots, X_r \cup S_1^2, Y_1 \cup S_1^2, \dots, Y_q \cup S_1^2)$ is a path-decomposition of $G[A_1^1]$ with path-width $\leq \alpha^2 n + f(\alpha n) - 1$. Similarly, we partition $G[B_1^1]$ into A_2^2, B_2^2, S_2^2 and do a similar procedure to get a path-decomposition of $G[B_1^1]$ with path-width $\leq \alpha^2 n + f(n) - 1$. We can now obtain a path-decomposition of G by merging these two decompositions by taking the union with S_1^1 on each bag. This new path-decomposition has path-width $\leq \alpha^2 n + f(\alpha n) + f(n) - 1$.

In the next step we partition each vertex set $A_1^2, B_1^2, A_2^2, B_2^2$ into A_i^3, B_i^3, S_i^3 and build up the path-decomposition from there. We continue this process recursively, building up the path-decompositions by merging. At each step k we achieve a path-width of $\leq \alpha^k n + \sum_{i=0}^{k-1} f(\alpha^i n) - 1$. We stop the algorithm when $\alpha^{k+1} n < 1$. Hence we get the bound

$$\text{pw}(G) \leq \sum_{i=0}^{\lfloor \log_{1/\alpha}(n) \rfloor + 1} f(\alpha^i n) \leq (\lfloor \log_{1/\alpha}(n) \rfloor + 1)f(n)$$

If $f(n) = \beta n^c$ then we can get the asymptotically better bound

$$\text{pw}(G) \leq \sum_{i=0}^{\lfloor \log_{1/\alpha}(n) \rfloor + 1} f(\alpha^i n) \leq \beta n^c \sum_{i=0}^{\lfloor \log_{1/\alpha}(n) \rfloor + 1} \alpha^{ic} \leq \beta n^c \frac{1}{1 - \alpha^c}$$

$\square$

Note that if G has an α -separator as described of order at most k then we get $\text{pw}(G) \leq k \log n$. Our goal now is to get rid of the dependence on n .

Robertson and Seymour [17] showed that path-width can be bounded by constants by excluding forests under the minor relation. We will need the following stronger theorem, first proven by Bienstock, et al. [18].

Theorem 3.9. *For every forest F , every graph with no minor isomorphic to F has path-width at most $|F| - 2$.*

We also need the following, which establishes that the order of α -separators is closed under minors.

Lemma 3.10. *Let G be a graph and H be a minor of G . If for every weighting of G there exists an α -separator of G of order $\leq k$, then for every weighting of H there exists an α -separator of H of order $\leq k$.*

Proof. Fix a weighting w_H of H . If H is a subgraph of G then the result follows by creating a weighting w_G of G such that if $v \in V(H)$ then $w_G(v) := w_H(v)$, and $w_G(v) := 0$ otherwise. If S is an α -separator of G into A, B, S then it follows that $S \cap V(H)$ is an α -separator of H into the partition $A \cap V(H), B \cap V(H), S \cap V(H)$ of $V(H)$. So we may assume that H is formed from G only by vertex contractions.

For each vertex $v \in H$, let M_v be the set of vertices in G who were contracted to become v (if v did not undergo contraction, $M_v = \{v\}$). For each $u \in M_v$ and for each v , define the weighting w_G of G as $w_G(u) := w_H(v)/|M_v|$. Let S_G be an α -separator of G of order $\leq k$ into a partition A_G, B_G, S_G of $V(G)$. Let

$$S_H := \{v \in V(H) \mid \exists u \in M_v \text{ s.t. } u \in S_G\},$$

$$A_H := \{v \in V(H) \mid u \in A_G \ \forall u \in M_v\} \text{ and } B_H := \{v \in V(H) \mid u \in B_G \ \forall u \in M_v\}$$

We claim that S_H is an α -separator of H of order $\leq k$ into the partition $A_H \cup B_H \cup S_H = V(H)$. The fact that $|S_H| \leq k$ is clear. Note that if a vertex $v \in V(H)$ is the result of contractions of vertices from both A_G and B_G then it must also result from the contraction of vertices from S_G , since S_G separates A_G and B_G and contractions are permitted only between vertices sharing an edge. This shows that A_H, B_H, S_H partition $V(H)$. Moreover, S_H separates A_H and B_H since if there is a path from A_H to B_H not going through S_H then there is an edge with one end in A_H and the other in B_H , meaning that the same is true for A_G and B_G , a contradiction.

Finally, by our construction of w_G it is easy to see that

$$w_H(A_H) \leq w_G(A_G) \leq \alpha w_G(V(G)) = \alpha w_H(V(H))$$

and similarly for $w_H(B_H)$. □

From these, we can derive the following unexpected bound of path-width.

Theorem 3.11. *Let G be a graph such that for every weighting of G there exists a $\frac{1}{2}$ -separator of order $\leq k$. Then $pw(G) \leq 3^{2k+o(k)}$.*

Proof. By theorem 3.9 it suffices to show that G does not contain a minor isomorphic to the complete rooted ternary tree T on n vertices, where we choose n is minimal such that $n \geq 3^{2k+o(k)}$. Suppose for contradiction that G contains a minor isomorphic to the complete rooted ternary tree on n vertices. Then by lemma 3.10, there exists a $\frac{1}{2}$ -separator of T of order $\leq k$. But this contradicts theorem 2.4, which states that the separator of T must be of order

$$\geq \frac{\log n}{2 \log 3} - \frac{\log n}{\log \log n} = \frac{\log 3^{2k+o(k)}}{2 \log 3} - \frac{\log 3^{2k+o(k)}}{\log \log 3^{2k+o(k)}} = k + \frac{o(k)}{2} - \frac{(2k + o(k)) \log 3}{\log((2k + o(k)) \log 3)} > k$$

for appropriate choice of $o(k)$. We can see this clearly if we let $o(k) = \frac{k}{C\sqrt{\log k}}$, for example (for large enough constant C). $\square$

We remark that if theorem 2.5 is used in place of theorem 2.4, then a bound on path-width of $O(2^{3^k})$ is obtained.

We have established a dependence between the order of $\frac{1}{2}$ -separators and the path-width of G . It seems possible that this dependence could be linear, as we have shown is the case between the order of $\frac{2}{3}$ -separators and tree-width of G .

Conjecture 1. *Let G be a graph such that for every weighting of G there exists a $\frac{1}{2}$ -separator of G of order $\leq k$. Then $\text{pw}(G) \leq O(k)$.*

4 Directed Graphs

In this section we generalize the notions of separators and tree-decompositions to digraphs. As we will see, dealing with digraphs is a lot more difficult than dealing with undirected graphs, but it is still possible to obtain some results reminiscent of those we discussed in undirected graphs.

Let D be a digraph. We say that S *separates* an ordered pair of vertex sets $A, B \subset V(D - S)$ if no edge has its tail in A and head in B . We say that S *weakly separates* an ordered pair of vertex sets $A, B \subset V(D - S)$ if every walk between two vertices in A that goes through a vertex in B also goes through a vertex in S . We say that S *symmetrically weakly separates* an (unordered) pair of vertex sets $A, B \subset V(D - S)$ if both A, B and B, A are weakly separated. A *separation* of D is an ordered pair of vertex subsets (A, B) of $V(D)$ such that $A \cap B$ separates $A \setminus B$ and $B \setminus A$, and $A \cup B = V(D)$. Its order is $|A \cap B|$. We define *weak separation* and *symmetric weak separation* analogously.

Let D be a weighted digraph with weighting w . For $\frac{1}{2} \leq \alpha < 1$ a α -separator of D is a set $S \subseteq V(D)$ such that there is a partition $A \cup B \cup S = V(D)$ with $w(A), w(B) \leq \alpha w(V(D))$ that separates A and B . We define the notions of an α -weak separator and α -symmetric weak separator analogously. A *balanced separator* of D is a set $S \subseteq V(D)$ such that each strongly connected component $C \subseteq D - S$ satisfies $w(V(C)) \leq \frac{1}{2}w(V(D))$.

It is clear that α -separators are also symmetric weak α -separators, which in turn are also weak α -separators. It is also easy to see that weak α -separators are not necessarily balanced separators if $\alpha > \frac{1}{2}$ but they are balanced separators if $\alpha = \frac{1}{2}$ (and hence the same follows for symmetric weak α -separators and α -separators). Lemmas 4.4 and 4.5 provide information on what strength of separators balanced separators imply.

We say that a directed graph D is a *directed acyclic graph* (DAG) if it contains no directed cycles. A *topological sort* of a directed graph D is a linear ordering of its vertices such that from every directed edge from $v \in V(D)$ to $w \in V(D)$, v comes before w in the linear ordering. The following theorem is elementary but very useful.

Theorem 4.1. *A directed graph D is a DAG if and only if it admits a topological sort.*

Lemma 4.2. *Let D be a weighted DAG with weighting w . If all vertices $v \in V(D)$ satisfy $w(v) \leq (2\alpha - 1)w(V(D))$ then $S = \emptyset$ is an α -separator of D , where $\frac{1}{2} < \alpha < 1$.*

Proof. Topologically sort the vertices in D into $v_1, \dots, v_n$. Add vertices to B consecutively in order right until $w(B) \geq (1 - \alpha)w(V(D))$. Let v_k be the last vertex assigned to B . After we are done adding

vertices to B , observe that $w(B) \leq \alpha w(V(D))$ since each vertex adds weight at most $(2\alpha - 1)w(V(D))$ (and $(1 - \alpha) + (2\alpha - 1) = \alpha$). Let $A = V(D - B)$. Then $w(A) < \alpha w(V(D))$ by our choice of B . Finally, observe that no vertex in $v_{k+1}, \dots, v_n$ has an edge pointing to a vertex in $v_1, \dots, v_k$ by definition of topological sort so A, B is the required partition. $\square$

Lemma 4.3. *Let D be a weighted DAG. If all vertices $v \in V(D)$ satisfy $w(v) \leq \frac{2}{3}w(V(D))$ then $S = \emptyset$ is a weak $\frac{2}{3}$ -separator of D .*

Proof. If some vertex v satisfies $w(v) \geq \frac{1}{3}w(V(D))$ let $A = \{v\}$ and $B = V(D - v)$. Clearly, $w(A), w(B) \leq \frac{2}{3}w(V(D))$. Since D contains no directed cycles then there is no closed walk in D and so in particular there is no walk between two vertices in A with a vertex in B , since A contains only one vertex. So A, B is the desired ordered pair of vertex sets.

Otherwise, we may assume that $w(v) < \frac{1}{3}w(V(D))$ for all $v \in V(D)$. Hence, we may apply lemma 4.2 with $\alpha = \frac{2}{3}$ to get the result. $\square$

Lemma 4.4. *Let D be a weighted digraph. If S is a balanced separator of D then S is a $\frac{3}{4}$ -separator of D .*

Proof. Let D have weighting w_D and let S be a balanced separator of D . Let $C_1, \dots, C_k$ be the strong components of $D - S$. Observe that there is no directed cycle in $D - S$ with vertices in multiple different components, since otherwise, those components would be strongly connected when put together, contradicting their maximality.

We construct H as follows. Contract each of the connected components C_i into a single vertex v_i while maintaining the edges coming into and out of each component. Contract S into a single vertex s while maintaining the edges coming into and out of S from and to each component. Moreover, let w_H be a weighting of H such that the weight in w_H of each vertex in H is equal the weight in w_D to its corresponding component in w_D , and the weight of s in w_H is equal to the weight of S in w_D .

Hence, we get that $w_H(v_i) = w_D(V(C_i)) \leq \frac{1}{2}w_D(V(D)) = \frac{1}{2}w_H(V(H))$ for all $1 \leq i \leq k$. Observe that $H - s$ is a DAG since there is no directed cycle between the vertices in $H - s$ (since there is no directed cycle between the corresponding components in $D - S$). Hence, if we let H' be a subgraph of H obtained removing all the edges incident on s then H' is a DAG.

We can now apply lemma 4.2 with $\alpha = 3/4$ to get that $\emptyset$ is a $\frac{3}{4}$ -separator of H' . Thus, s is a $\frac{3}{4}$ -separator of H . Thus, S is a $\frac{3}{4}$ -separator of D . $\square$

The following is proven very similarly.

Lemma 4.5. *Let D be a weighted digraph. If S is a balanced separator D then S is a weak $\frac{2}{3}$ -separator of D .*

4.1 Directed Tree-width

We will deal with two notions of directed tree-decompositions. The first was introduced by Johnson, et al. [19]. An *arborescence* is a directed tree with a root r such that for every other vertex v there exists one directed path from r to v . A *NW-directed tree-decomposition* of a digraph D is a triple $\mathcal{T} = (T, \beta, \gamma)$ where T is an arborescence, $\beta : V(T) \rightarrow 2^{V(D)}$ and $\gamma : E(T) \rightarrow 2^{V(D)}$ (so each vertex and edge in T maps to a vertex set in D) such that

(NW1) $\{\beta(t) : t \in T\}$ is a partition of $V(D)$ into nonempty sets.

(NW2) If $st = e \in E(T)$ then γ_e weakly separates $\beta(T_t) \setminus \gamma_e$ and $V(D - (\beta(T_t) \cup \gamma_e))$.

We remark that NW stands for no walk. Observe that if $e = (s, t)$ then $\beta(T_t)$ is the union of the vertex set of some strong components in $D - \gamma_e$.

The following notion of a directed tree-decomposition was defined by Kawarabayashi and Kreutzer [20]. It is useful because, unlike the NW-directed tree-decomposition, it is closed under *butterfly minors* and is bounded when excluding a *directed grid* (we will not elaborate on this, see [20], [21]). A $\text{NCW}_\emptyset$ -directed tree-decomposition of a digraph D is a triple $\mathcal{T} = (T, \beta, \gamma)$ as above such that

(NCW $_\emptyset$ 1) $\{\beta(t) : t \in T\}$ is a partition of $V(D)$ into (potentially empty) sets.

(NCW $_\emptyset$ 2) If $st = e \in E(T)$ then there is no closed walk in $D - \gamma_e$ containing a vertex of $\beta(T_t)$ and a vertex of $V(D - \beta(T_t))$.

We remark that $\text{NCW}_\emptyset$ stands for no closed walk, where $\emptyset$ emphasizes the fact that some of the bags may be empty. Observe that if $e = st$ then $\beta(T_t) \setminus \gamma_e$ is the union of the vertex set of some strong components in $D - \gamma_e$.

For both definitions we call β *bags* and γ *guards*. We write $e \sim t$ if an edge e is incident on a vertex t . For both definitions we let $\Gamma_t = \beta_t \cup \bigcup_{e \sim t} \gamma_e$. We say that the *directed tree-width* of a graph is the minimum over the directed tree-decompositions $\mathcal{T} = (T, X_t)$ of $\max_{t \in T} |\Gamma_t| - 1$. There are multiple other notions of directed tree-width which we omitted here. For an extensive discussion of different notions of directed tree-decompositions and their relations see [22].

It is not hard to prove that the width of a $\text{NCW}_\emptyset$ -directed tree-decomposition is at most the width of a NW-directed tree-decomposition. As we will see, both of these behave similarly with respect to some notions of directed separators.

The following result was essentially first given by Johnson, et al. [19], but was stated in terms of *havens* and shown for NW-directed tree-decompositions. We provide a constructive proof that works for every weighting.

Theorem 4.6. *Let D be a weighted digraph. If D has an $\text{NCW}_\emptyset$ -directed tree-decomposition of width $< k$ then there exists a balanced separator of D of order $\leq k$.*

Proof. Let D have weighting w and let $\mathcal{T} = (T, X_t)$ be a $\text{NCW}_\emptyset$ -directed tree-decomposition of width $< k$. Let r be the root of T . If every child c of r satisfies $w(\beta(T_c)) \leq \frac{1}{2}w(V(G))$ then we let $S := \Gamma_r$. Since Γ_r contains the guard of each c and each $\beta(T_c) \setminus \gamma_e$ where $sc = e \in E(T)$ is the union of the vertex set of strongly connected components (which are distinct for distinct c since $\beta(t)$ for $t \in T$ partitions $V(D)$) then S is a balanced separator and we are done.

Otherwise, there is some unique c such that $w(\beta(T_c)) > \frac{1}{2}w(V(G))$. We iterate this procedure on the children of c , redefining what c is, right until every child c' of c satisfies $w(\beta(T_{c'})) \leq \frac{1}{2}w(V(G))$. In this case we let $S := \Gamma_c$. Observe that $(\bigcup\{\beta(t) : t \notin T_c\}) \setminus S$ has weight $< \frac{1}{2}w(V(G))$ since $w(\beta(T_c)) > \frac{1}{2}w(V(G))$. Moreover, since Γ_c contains the guard of each c' and each $\beta(T_{c'}) \setminus \gamma_e$ such that $sc' = e \in E(T)$ is the union of the vertex set of strongly connected components (which are distinct for distinct c since $\beta(t)$ for $t \in T$ partitions $V(D)$), then S is a balanced separator and we are done. $\square$

As was the case with undirected tree-width, it is possible to get a partial converse of theorem 4.6. The following was first given by Johnson, et al. [19], albeit stated in terms of havens.

Theorem 4.7. *Let D be a graph such that for every 0-1 weighting of D there exists a balanced separator of order $\leq k$. Then D has a NW-directed tree-decomposition of width $\leq 3k + 1$.*

The following result is slightly stronger than theorem 4.7 in terms of the separator used, but weaker in the constant obtained. It follows a very similar proof strategy as theorem 3.5.

Theorem 4.8. *Let D be a graph such that for every 0-1 weighting of D there exists a symmetric weak $\frac{2}{3}$ -separation of order $\leq k$. Then D has a NW-directed tree-decomposition of width $< 4k$.*

Proof. We consider the following recursive algorithm. It takes as input a digraph D , a symmetric weak separation (X, Y) with $W := X \cap Y$ where $|W| \leq 3k$, and a NW-directed tree-decomposition $\mathcal{T} = (T, \beta, \gamma)$ of $D[Y]$ of width at most $4k$ in which some $\beta_t \cup \gamma_e$ contain all vertices in $W := X \cap Y$, where e is an edge with its head in t .

If $X \subseteq W$ then $X \subseteq Y$, so $\mathcal{T}$ is a NW-directed tree-decomposition of D and we are done. So we may assume that $X \not\subseteq W$. If $|X| \leq 3k$ then we create a leaf t' in T with an edge e' pointing from t to t' where $\gamma(e') = W$ and $\beta(t') = X \setminus W$ and we are done. So we may assume that $|X| \geq 3k + 1$. Since $|W| < 3k + 1$ then we can take vertices from $X \setminus Y$ and add them to W to create a new vertex set W^* , where $|W^*| = 3k + 1$.

To extend $\mathcal{T}$, we let D have weighting w which sets all vertices in W^* to 1 and all other vertices in D to 0. Let (A', B') be a symmetric weak separation of D such that $S' := A' \cap B'$ is a symmetric weak $\frac{2}{3}$ -separation of D of order $\leq k$. Let $(A, B) := (A' \cap X, B' \cap X)$ and $D[X]$ have weighting w (where w is restricted to the vertices in X). Then (A, B) is a symmetric weak separation of $D[X]$ such that $S := A \cap B$ is a symmetric weak $\frac{2}{3}$ -separator of $D[X]$ of order $\leq k$.

If $S \subseteq W$ then we don't create a new leaf in T and we let $t' := t$ and $e' := e$. Otherwise, we may assume $S \not\subseteq W$. In this case, create a leaf t' in T with an edge e' pointing from t to t' . Let $\gamma_{e'} := W$ and $\beta_{t'} := S \setminus W$ (note that $\beta(t') \neq \emptyset$ since $S \not\subseteq W$). Observe that $|\Gamma_t| = |\beta_t \cup \bigcup_{e \sim t} \gamma_e| \leq 3k$ since $\beta_t \cup \gamma_e$ already contained W . Moreover, $|\Gamma_{t'}| = |\beta_{t'} \cup \bigcup_{e \sim t'} \gamma_e| \leq |W| + |S| \leq 4k$. So the width of the tree-decomposition is still less than $4k$.

Let $X_A := A, Y_A = Y \cup S, D_A := D[X_A \cup Y_A]$ and $W_A = X_A \cap Y_A$. Observe that $|W_A| \leq |W_A \setminus B| + |S| \leq |W^* \setminus B| + |S| \leq \frac{2}{3}|W^*| + k < 3k + 1$. Moreover, $\beta_{t'} \cup \gamma_{e'}$ contains $W_A \subseteq W \cup S$. So we can call the algorithm on $D_A, (X_A, Y_A), \mathcal{T}$. This way, the algorithm extends $\mathcal{T}$ to D_A .

Now, we let $X_B := B, Y_B = Y \cup A, D_B := D[X_B \cup Y_B] = D$ and $W_B = X_B \cap Y_B$. Observe that $|W_B| \leq |W_B \setminus A| + |S| \leq |W^* \setminus A| + |S| \leq \frac{2}{3}|W^*| + k < 3k + 1$. Moreover, $\beta_{t'} \cup \gamma_{e'}$ contains $W_B \subseteq W \cup S$. So we can call the algorithm on $D_B, (X_B, Y_B), \mathcal{T}$. This way, the algorithm extends $\mathcal{T}$ to $D_B = D$.

To initialize this algorithm, we let $Y \subseteq V(D)$ be an arbitrary vertex set of size $3k$ and $X = V(D)$, and let $\mathcal{T} = (T, \beta, \gamma)$ have one root vertex r and bag $\beta_r = X \cap Y = Y$ (if $|V(D)| < 3k$ then the claim follows trivially). Observe that this algorithm terminates since at each step it either doesn't recurse or we have $|W^*| = 3k + 1$ and $|S| \leq k$, meaning that $A, B \subsetneq X$ (indeed if, say, $A = X$ then $w(A \setminus B) = w(X) - w(S) \geq 2k + 1 > \frac{2}{3}(3k + 1) = \frac{2}{3}w(X)$ which contradicts the fact that S is a symmetric weak $\frac{2}{3}$ -separation of $D[X]$). This means that each time we call the algorithm recursively, X gets smaller, and so we will eventually get $|X| \leq 3k$ which is the stopping condition of the recursion. $\square$

We remark that this theorem can be generalized to symmetric weak α -separators where $\frac{1}{2} \leq \alpha < 1$. In this case the width is less than $\frac{2-\alpha}{1-\alpha}$, and in each step of the algorithm we maintain a separation (X, Y) of order at most $\frac{1}{1-\alpha}$. This way, one can check that if $\alpha = \frac{3}{4}$ then we get width less than $5k$.

4.2 Planar Digraphs

We will now move to a different notion of width, which can be viewed as a generalization of *branch-width* (see [23]) that only works on planar digraphs, and its relation to a different type of separator.

Graphs in this section may have parallel edges but not loops. Let Σ denote the 2-sphere and assign the orientation clockwise to Σ . A *didrawing* is a planar drawing G on σ such that its edges are directed. If G is a digraph, we say a digraph H is a *subdivision* of G if H can be obtained from G by repeatedly replacing an edge uv with a new vertex w and two edges uw, wv (u, v might be equal). Let us say a digraph G *embeds* a digraph H if there is a subdigraph of G that is isomorphic to a subdivision of H . Similarly, a didrawing H is a Σ -*subdivision* of a didrawing G if H can be obtained from G by repeatedly replacing an edge uv with a new vertex w and two edges uw, wv , such that $uw \cup wv = uv$. A didrawing G Σ -*embeds* a didrawing H if there is a subdidrawing of G that is homeomorphic to a Σ -subdivision of H .

A tree T is *ternary* if each of its vertices has degree one or three. Let S be a finite set with $|S| \geq 2$, let T be a ternary tree, and let τ be a bijection from S onto the set of leaves of T . We call (T, τ) a *carving* of S . For each edge e of T let $T_1(e), T_2(e)$ be the two components of $T - e$. We call

$$(\{v \in S : \tau(v) \in V(T_1(e))\}, \{v \in S : \tau(v) \in V(T_2(e))\})$$

and its reverse the *partitions of S corresponding to e under (T, τ)* .

Let G be a didrawing. A *dart* of G is a pair (v, e) where $v \in V(G)$, $e \in E(G)$ and e is incident with v . Let F be a simple closed curve in Σ , such that F has at most one point in common with each edge, and for each region r , either $F \cap r = \emptyset$ or $F \cap r$ is the interior of a line. We call such a curve F *good*. Let F be good, and let Δ_1, Δ_2 be the two closed discs in Σ bounded by F . We say a dart (v, e) *belongs to Δ_i* if either v belongs to the interior of Δ_i , or $v \in F$ and $e \setminus \{v\}$ is a subset of the interior of Δ_i . For $i = 1, 2$, let D_i be the set of darts that belong to Δ_i ; we call (D_1, D_2) and (D_2, D_1) the *dart-partitions from F* . A partition (A, B) of the set of all darts is *sensible* if (A, B) is a dart-partition from F for some F -separator.

Let F be a good curve, and let $\Delta_1, \Delta_2, D_1, D_2$ be as before. A *change-region* for F is a region r incident with two edges e, f that both cross F (since F is good, there cannot be three edges crossing F and incident with r), such that Δ_1 contains the head of one of e, f and Δ_2 contains the head of the other. (If r, e, f are as above, it follows that $F \cap (r \cup e \cup f)$ is a line with ends in the interiors of e and f). The number of change-regions for F is its *change number*. Let us say the *cost* of (D_1, D_2) is the sum of the change number of F and $|F \cap V(G)|$.

We say a didrawing has *dart-width* at most k if there is a carving (T, τ) of the set of all darts, such that for each edge e of T , the corresponding dart-partition is sensible and has cost at most k . A digraph is *weakly 2-edge-connected* if its undirected counterpart is 2-edge-connected (no didrawing with a cut-edge admits a sensible dart-partition).

Let G be a weighted didrawing with weights w . For a set of darts D , we let $V(D)$ be the set of vertices in the darts of D . A *balanced dart-separator* of G of order k is a good curve F of cost k such that if (D_1, D_2) is the dart-partition from F then $w(V(D_1 \setminus D_2)), w(V(D_2 \setminus D_1)) \leq \frac{2}{3}w(V(G))$.

Theorem 4.9. *Let G be a weighted loopless 2-edge-weak didrawing. If G has dart-width at most k then it has a balanced dart-separator of order $\leq k$.*

Proof. Let (T, τ) be a carving of G such that for each edge e of T the corresponding dart-partition has cost at most k . We consider the following algorithm. Pick some edge $e \in T$ and consider its

corresponding dart-partition (D_1, D_2) from some good curve F . If $w(V(D_1 \setminus D_2)), w(V(D_2 \setminus D_1)) \leq \frac{2}{3}w(V(G))$ then F is the desired balanced dart-separator. Otherwise, suppose without loss of generality that $w(V(D_1 \setminus D_2)) > \frac{2}{3}w(V(G))$. In this case we have $w(V(D_2)) < \frac{1}{3}w(V(G))$.

Let $T_1(e)$ be the subtree of T associated with D_1 . Observe that there exist two edges $e', e'' \in T_1(e)$ that share a vertex with e . Indeed, $T_1(e)$ cannot be a single vertex since that would mean that D_1 is composed of a single dart, say (v, f) . But since G is 2-edge-weak then there must be at least one other dart containing v in D_2 . Hence, $w(V(D_1 \setminus D_2)) = w(\emptyset) = 0$, a contradiction.

Let (D'_1, D'_2) be the corresponding dart-partition of e' from some good curve F' and (D''_1, D''_2) the corresponding dart-partition of e'' from some good curve F'' . Without loss of generality say we have $D'_1, D''_1 \subset D_1$. Then, D'_1, D''_1 partition D_1 , and also D'_1, D_2 partition D'_2 . We have several cases

1. Suppose $w(V(D'_1 \setminus D'_2)) > \frac{2}{3}$. Then redefine $e := e'$ and go to the second paragraph of the proof.
2. Suppose $w(V(D'_2 \setminus D'_1)) > \frac{2}{3}$. Then $w(D'_1) < \frac{1}{3}w(V(G))$. Since we also have $w(V(D_2)) < \frac{1}{3}w(V(G))$ then $w(V(D'_2 \setminus D'_1)) < \frac{2}{3}w(V(G))$ (since D'_1, D_2 partition D'_2). In this case, we either have case 1 or case 3 but for the dart-partition (D''_1, D''_2) of e'' from F'' . So we redefine $e := e''$ and go to the second paragraph of the proof.
3. Suppose $w(V(D'_1 \setminus D'_2)) \leq \frac{2}{3}w(V(G))$ and $w(V(D'_2 \setminus D'_1)) \leq \frac{2}{3}$. Then F' is our desired balanced dart-separator.

This algorithm clearly terminates since it traverses the edges of T without going back to previous edges, and so it must eventually reach a leaf vertex (the algorithm must terminate before reaching a leaf vertex). $\square$

We now want to prove a partial converse of this result. For even k , a $k \times k$ *diwall* is a didrawing as pictured in Figure 1. It can be constructed by placing k disjoint directed paths P_i , for $1 \leq i \leq k$ horizontally, alternating in their direction (so odd-indexed paths go to the right and even-indexed paths go to the left), each of length $2k - 1$. Then, for every four vertices, between consecutive odd-and-even indexed paths, add an edge going up between the 1st pair of vertices and an edge going down between the 4th pair of vertices. Similarly, for every four vertices, between consecutive even-and-odd-indexed paths add an edge going up between the 2nd pair of vertices and an edge going down between the 3rd pair of vertices.

We shall need the following powerful result, proven by Chudnovsky and Seymour [24].

Theorem 4.10. *For all even $k \geq 2$, if G is a weakly 2-edge-connected didrawing that does not embed the $k \times k$ diwall, then G has dart-width at most $16k^2$.*

Leveraging the diwall's grid structure, and recalling the fact that grids have balanced separators of order $\Omega(\sqrt{n})$, as we showed in section 2, we prove the following.

Theorem 4.11. *Let G be the unweighted $k \times k$ diwall where $k \geq 12$, and k even. Then G has no balanced dart-separator of cost $< \frac{k}{4}$.*

Proof. Let H be an undirected graph constructed by placing a vertex in each region enclosed by an directed 8-cycle going clockwise, and placing an edge for each region enclosed by four edges that goes between these clockwise cycles. Then H is isomorphic to the $\frac{k}{2} \times \frac{k}{2}$ square grid graph. See Figure 1.

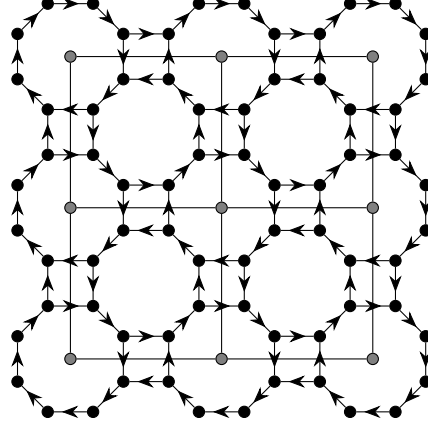


Figure 1: The 6×6 diwall in black. The 3×3 square grid in gray.

Observe that in G , the set of directed 8-cycles going clockwise form a partition of the vertices. Hence, each vertex in H corresponds uniquely to 8 vertices in G .

Claim: If there is a balanced dart-separator F of unweighted G with cost at most c then there is a $\frac{2}{3}$ -separator S of unweighted H of order at most c .

To see so, let (D_1, D_2) be the dart-partition from F . Now, assign a direction to F and start traversing the curve. Consider that every time F goes between regions in G that correspond, in H , to different vertices, different edges, or a vertex and edge that don't have an incidence relation, then the cost of F increases by at least 1.

Let S' be a set of vertices and edges in H , where the vertices correspond to the regions enclosed by directed 8-cycles going clockwise in G that F goes through and the edges correspond to the regions enclosed by 4 edges in G that F goes through, such that these edges don't already have either of their ends in S' . Hence, $|S'| \leq c$.

Observe that we can partition $V(H - S')$ into vertex sets A', B' such that if $v \in A'$ then v corresponds to a directed 8-cycle going clockwise in G whose darts belong entirely in Δ_1 (and hence its vertices are entirely in $V(D_1)$), and if $v \in B'$ then v corresponds to a directed 8-cycle going clockwise in G whose darts belong entirely in Δ_2 (and hence its vertices are entirely in $V(D_2)$). Notice that the vertices $v \in S'$ correspond to the regions enclosed by directed 8-cycles going clockwise in G which F intersects with. Hence, since F is a balanced dart-separator for unweighted G , then $|A'|, |B'| \leq (\frac{2}{3})(\frac{1}{8})|V(G)| \leq \frac{2}{3}|V(H)|$ and A', B' are separated in $H - S'$.

Therefore, we can let S be a vertex set obtained by adding all the vertices from S' and exchanging each of the edges S' for one of its two ends, if both ends are not already in S . Then clearly, S is a $\frac{2}{3}$ -separator of H of order at most c . This proves the claim.

By theorem 2.10 H has no $\frac{2}{3}$ -separator of order $< k/4$ and so by the earlier claim, G has no balanced dart-separator of cost $< k/4$. $\square$

Lemma 4.12. *Let G be a weakly 2-edge-connected didrawing that embeds a 2-edge connected didrawing H . If for every 0-1 weighting G has a balanced dart-separator of cost $\leq k$, then for every 0-1 weighting H has a balanced dart-separator of cost $\leq k$.*

Proof. Let w_H be a 0-1 weighting of H . Construct G out of H by subdividing H and adding new

vertices and edges (which is possible by the definition of embedding). Let w_G be a weighting of G that is equal to w_H on the vertices originally in H , and 0 on the rest of the vertices in $V(G)$. Let F be a balanced dart-separator of G of cost $\leq k$. Then we claim that F is also a balanced dart-separator of H of cost $\leq k$.

The fact that F results in a balanced dart-separator of H follows from our construction of w_G and the fact that if F is good in G then it is also good in H . To see that the cost is $\leq k$, observe that if F goes through a given vertex or edge in G created by a subdivision of an edge of H then F will go through the unsubdivided edge, so in this case the cost will either decrease or stay the same. And if F goes through a given vertex or edge in G that is not part of the subdivision of H , then it will not through any edge or vertex in H , so again in this case the cost will either decrease or stay the same. $\square$

Theorem 4.13. *Let G be a loopless weakly 2-edge-connected drawing with a balanced dart-separator of cost at most k for every 0-1 weighting, where $k \geq 12$. Then G has dart-width at most $64k^2$.*

Proof. By theorem 4.10 it is sufficient to show that G does not embed the $4k \times 4k$ diwall. Suppose for contradiction that G embeds the $4k \times 4k$ diwall. Then by lemma 4.12, the $4k \times 4k$ diwall has a balanced dart-separator of cost at most k . But this contradicts theorem 4.11. $\square$

It may be possible to obtain a linear dependence between the cost of a balanced dart-separator and dart-width. However, the usual proof strategy used in theorems 3.5 and 4.8 doesn't quite work in this case because of the nature of using a closed curve F as a separator, so it is unclear how to proceed.

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